

بِسْمِ اللَّهِ الرَّحْمَنِ الرَّحِيمِ

KHYBER PAKHTUNKHWA BOARD OF TECHNICAL EDUCATION

DIPLOMA IN INFORMATION TECHNOLOGY (DIT)
FINAL YEAR PROJECT THESIS

PHARMACY MANAGEMENT SYSTEM

Submitted in Partial Fulfillment of the Requirements for the Award of DIT Diploma

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CERTIFICATE OF APPROVAL

This is to certify that the project thesis titled 'PHARMACY MANAGEMENT SYSTEM' submitted by Muhammad Zahid (Roll No: 238762) has been thoroughly evaluated and approved as meeting the technical requirements for the Diploma in Information Technology (DIT).

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Head of Department

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ACKNOWLEDGEMENT

First and foremost, I express my deepest gratitude to Almighty Allah for giving me the strength and capability to complete this project. I would also like to express my sincere appreciation to my project supervisor, instructors, and family for their guidance, encouragement, and support throughout the development of this Microsoft Access database application.

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CHAPTER 1: INTRODUCTION & SYSTEM SCOPE

1.1 Background of the Project

Pharmaceutical retail outlets require precise inventory management, accurate expiration tracking, and rapid billing processing. Manual registration systems lead to errors, missing stock records, and slow customer checkout. The Pharmacy Management System (PMS) is developed to digitize and automate all pharmacy operations.

1.2 System Objectives

- Automate retail customer sales with single-receipt multi-item entry.
- Track stock levels, unit purchase prices, batch numbers, and expiration dates in real time.
- Manage supplier procurement and maintain vendor order histories.
- Generate automated executive summaries, financial reports, and customer bills.

1.3 System Flowchart

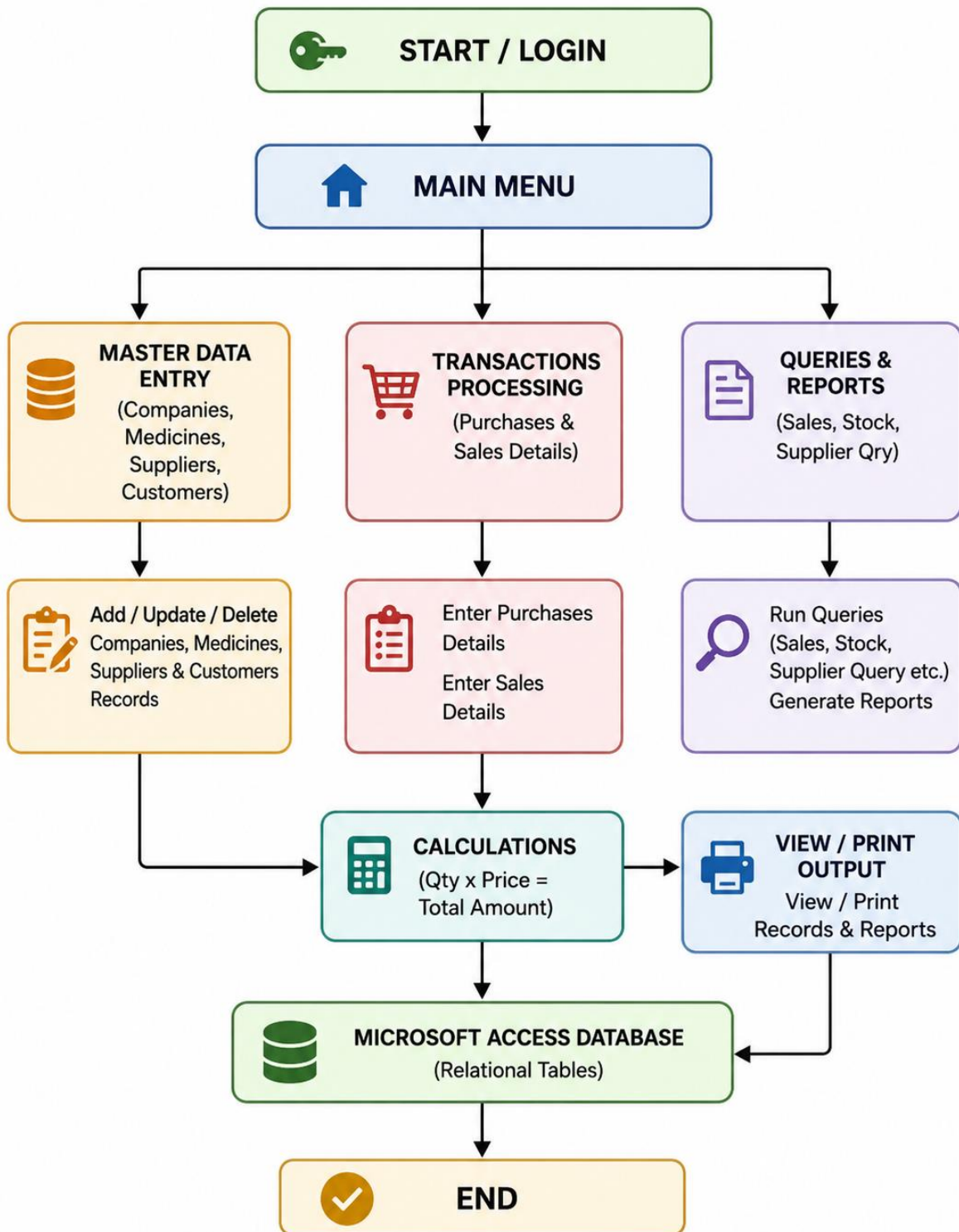


Figure 1: System Architecture & Data Flowchart Diagram

CHAPTER 2: SYSTEM ANALYSIS & ENVIRONMENT

The application is developed using Microsoft Office Access as the RDBMS engine. MS Access provides a reliable relational database, built-in SQL query engine, graphical form designer, and report generator.

Database Engine:	Microsoft Access (Jet/ACE Engine)
Query Language:	Structured Query Language (ANSI SQL)
User Interface:	MS Access Form Tools with Tabbed Interface
Reporting Tool:	MS Access Integrated Report Builder
Operating System:	Microsoft Windows 10

CHAPTER 3: DATABASE DESIGN & TABLES

The database consists of 7 normalized tables structured in 3rd Normal Form (3NF):

3.1 Companies Table

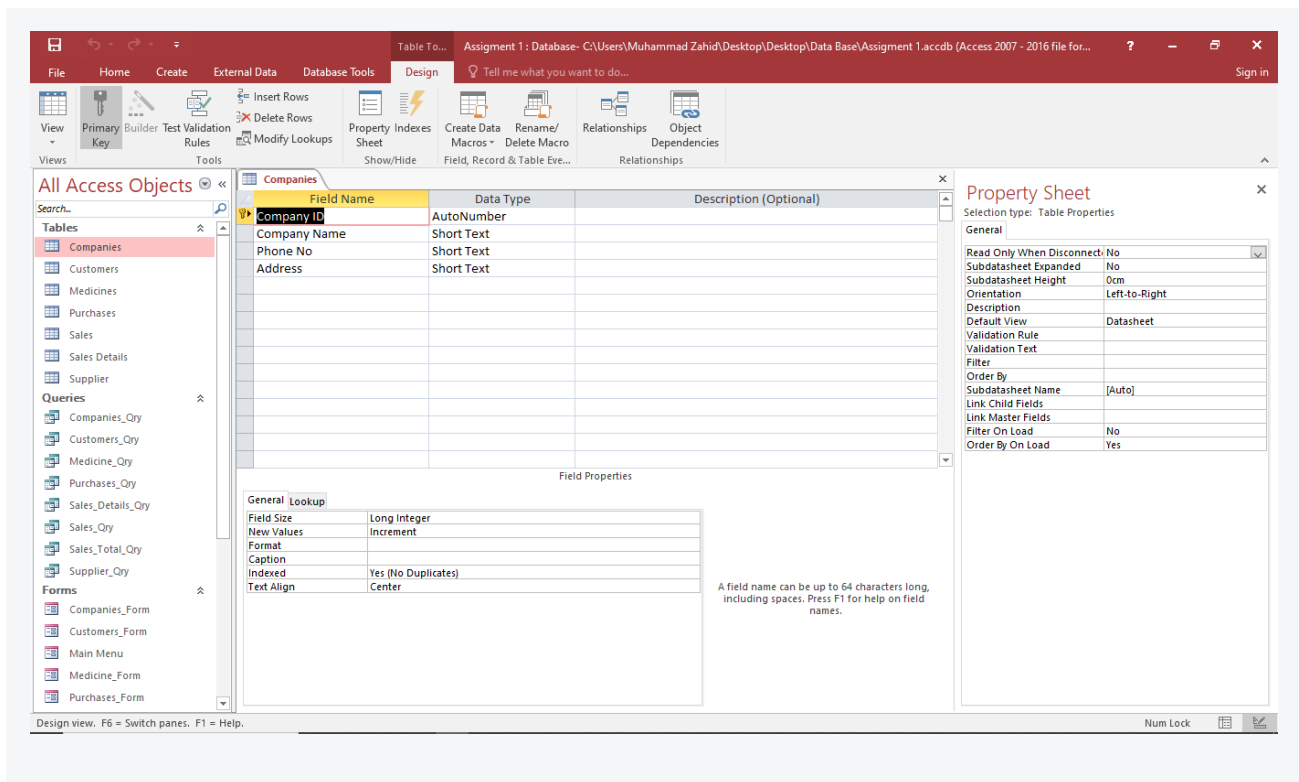


Figure 3: Companies Table Design & Datasheet View

3.2 Customers Table

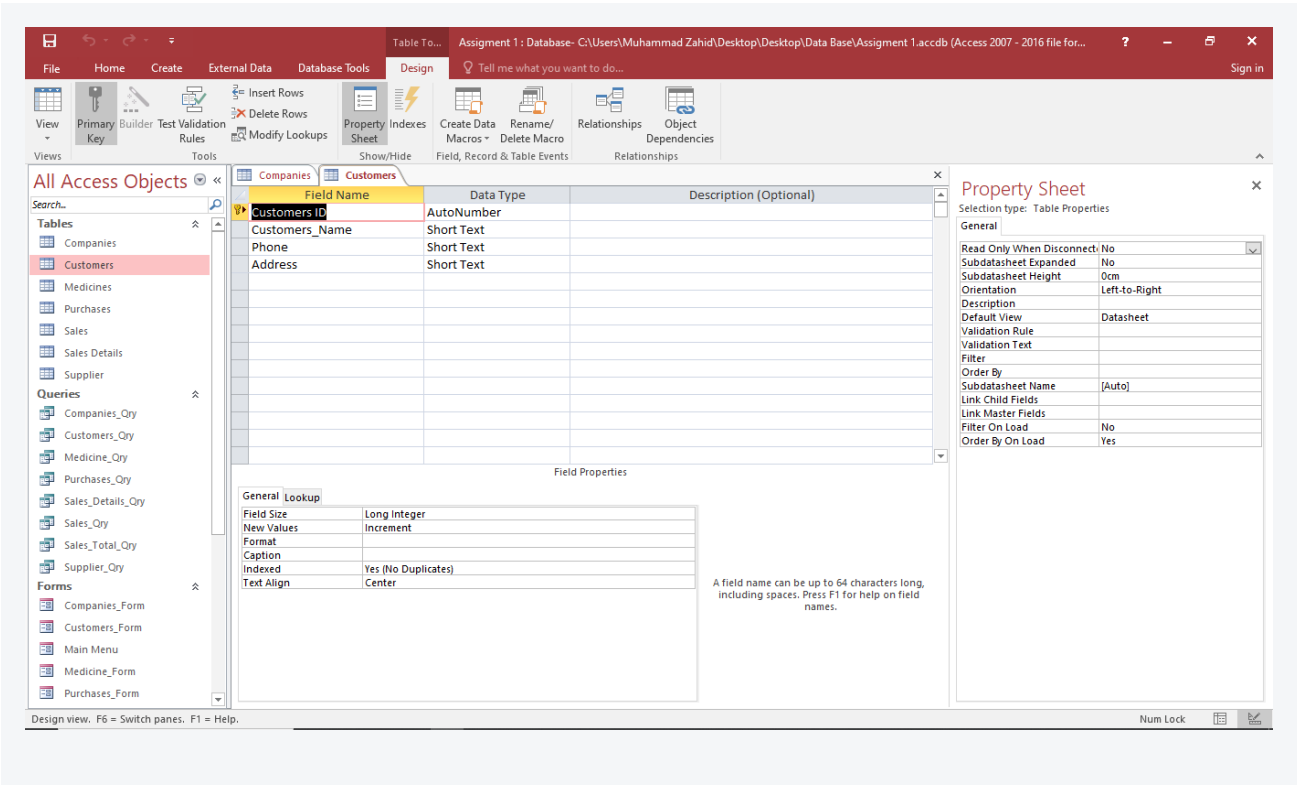


Figure 4: Customers Table Design & Datasheet View

3.3 Medicines Table

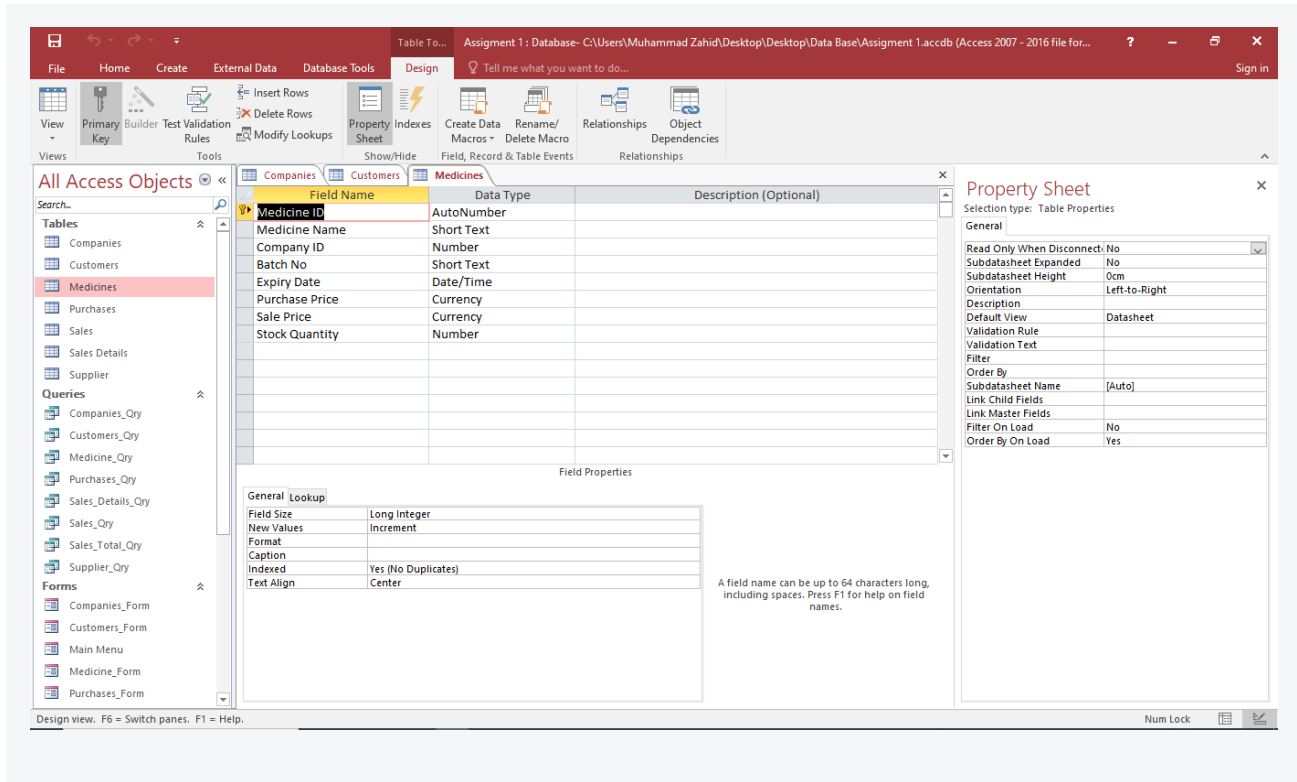


Figure 5: Medicines Inventory Table Design View

3.4 Sales & Sales Details Tables

The screenshot displays the Microsoft Access application window for a database named "Assignment 1 : Database- C:\Users\Muhammad Zahid\Desktop\Data Base\Assignment 1.accdb". The "Design" ribbon is active, showing the "Table Tools" group. The "Sales" table is selected in the "All Access Objects" pane on the left. The main area shows the "Sales" table in Design view, with the following fields:

Field Name	Data Type	Description (Optional)
Sale ID	AutoNumber	
Sale Date	Date/Time	
Customers ID	Number	

The "Property Sheet" on the right shows the "Table Properties" for the "Sales" table. The "General" tab is selected, displaying various properties such as "Read Only When Disconnected", "Subdatasheet Expanded", "Subdatasheet Height", "Orientation", "Description", "Default View", "Validation Rule", "Validation Text", "Filter", "Order By", "Subdatasheet Name", "Link Child Fields", "Link Master Fields", "Filter On Load", and "Order By On Load".

Below the table design, the "Field Properties" section is visible, showing the "General" tab for the "Sale ID" field. The properties are:

Property	Value
Field Size	Long Integer
New Values	Increment
Format	
Caption	
Indexed	Yes (No Duplicates)
Text Align	Center

A note at the bottom right states: "A field name can be up to 64 characters long, including spaces. Press F1 for help on field names."

Figure 6: Sales & Sales Details Table Schema

3.5 Purchases & Supplier Tables

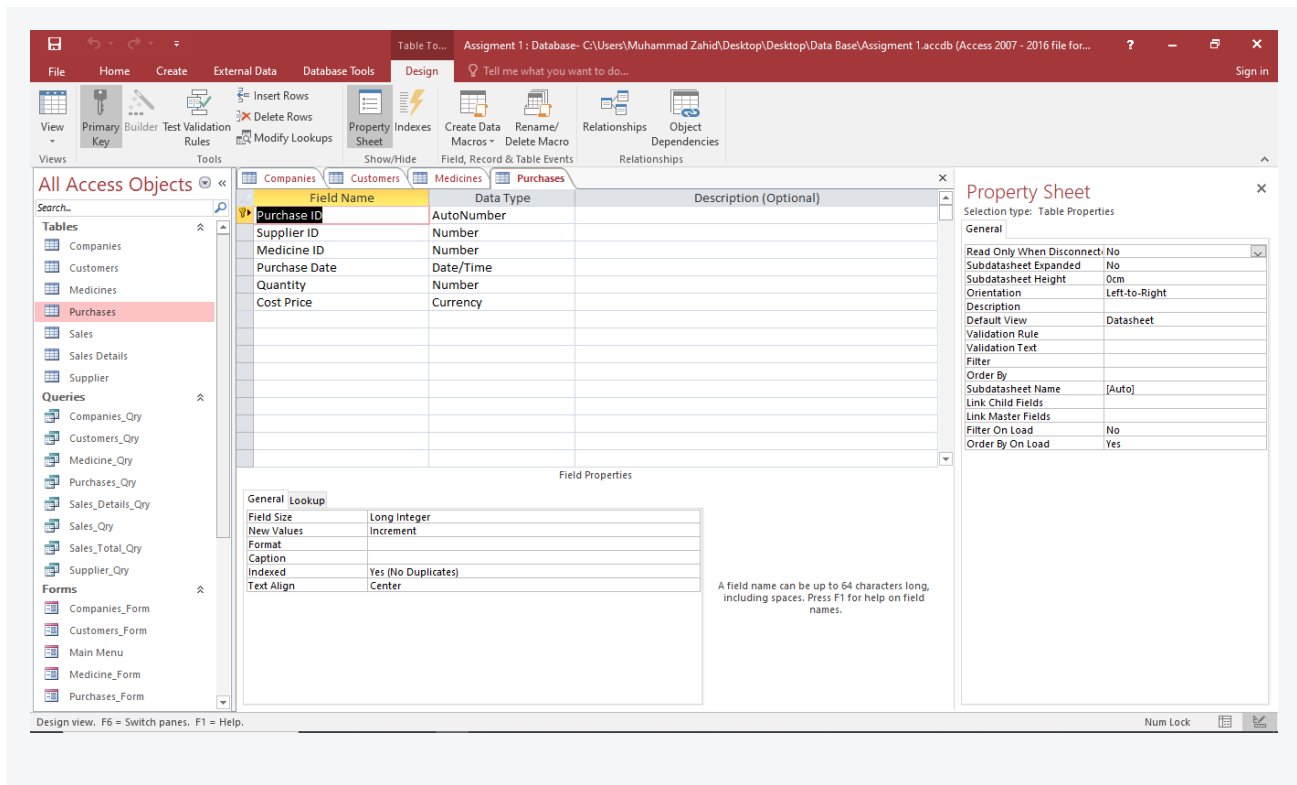


Figure 7: Purchases Table & Vendor Structure

CHAPTER 4: RELATIONAL MODEL & ER DIAGRAM

All tables in the system are linked through Primary Key - Foreign Key relationships with Enforce Referential Integrity enabled.

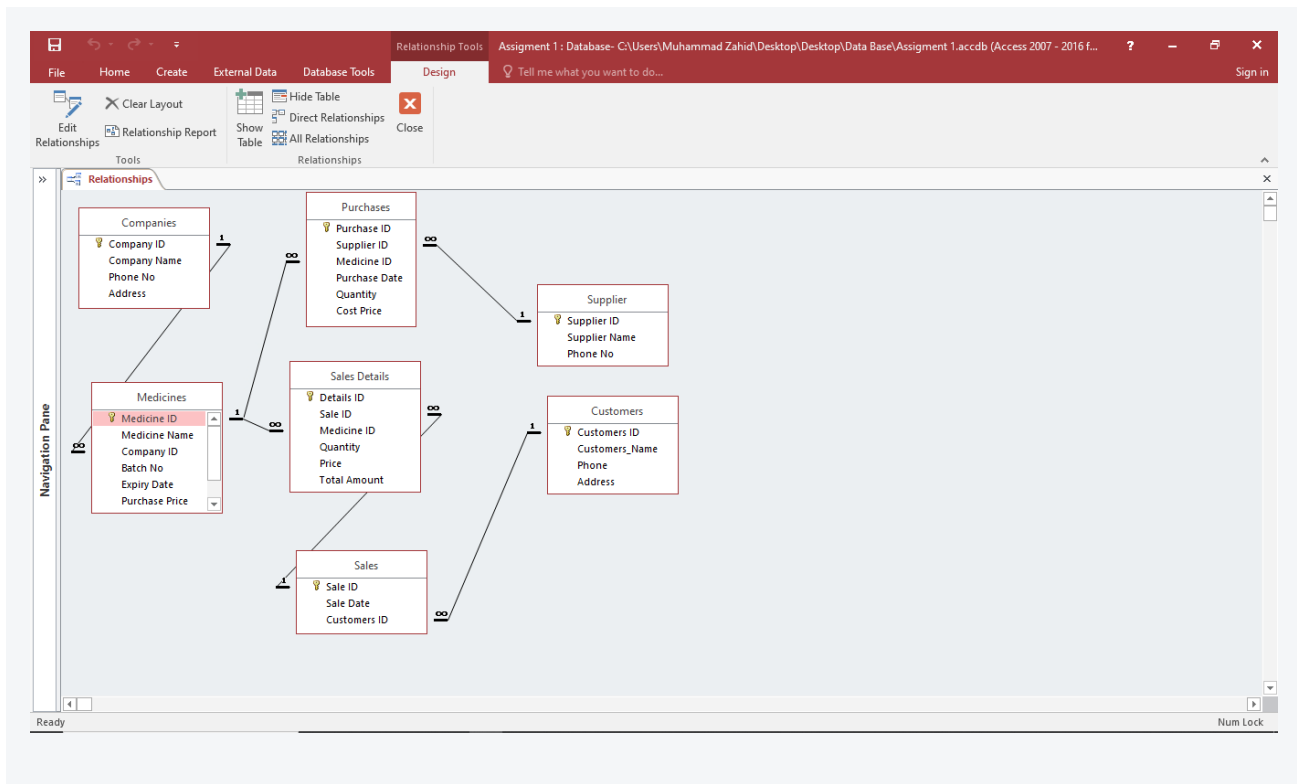


Figure 2: Complete Entity Relationship Diagram (MS Access Relationships Window)

CHAPTER 5: SQL QUERIES IMPLEMENTATION

The database contains 8 essential SQL queries driving calculation and dynamic views:

5.1 Sales Total Aggregation Query (Sales_Total_Qry)

SQL Code:

```
SELECT Sales.[Customers ID], Sum([Sales Details].[Total Amount]) AS [Total Amount] FROM
Sales INNER JOIN [Sales Details] ON Sales.[Sale ID] = [Sales Details].[Sale ID] GROUP BY
Sales.[Customers ID];
```

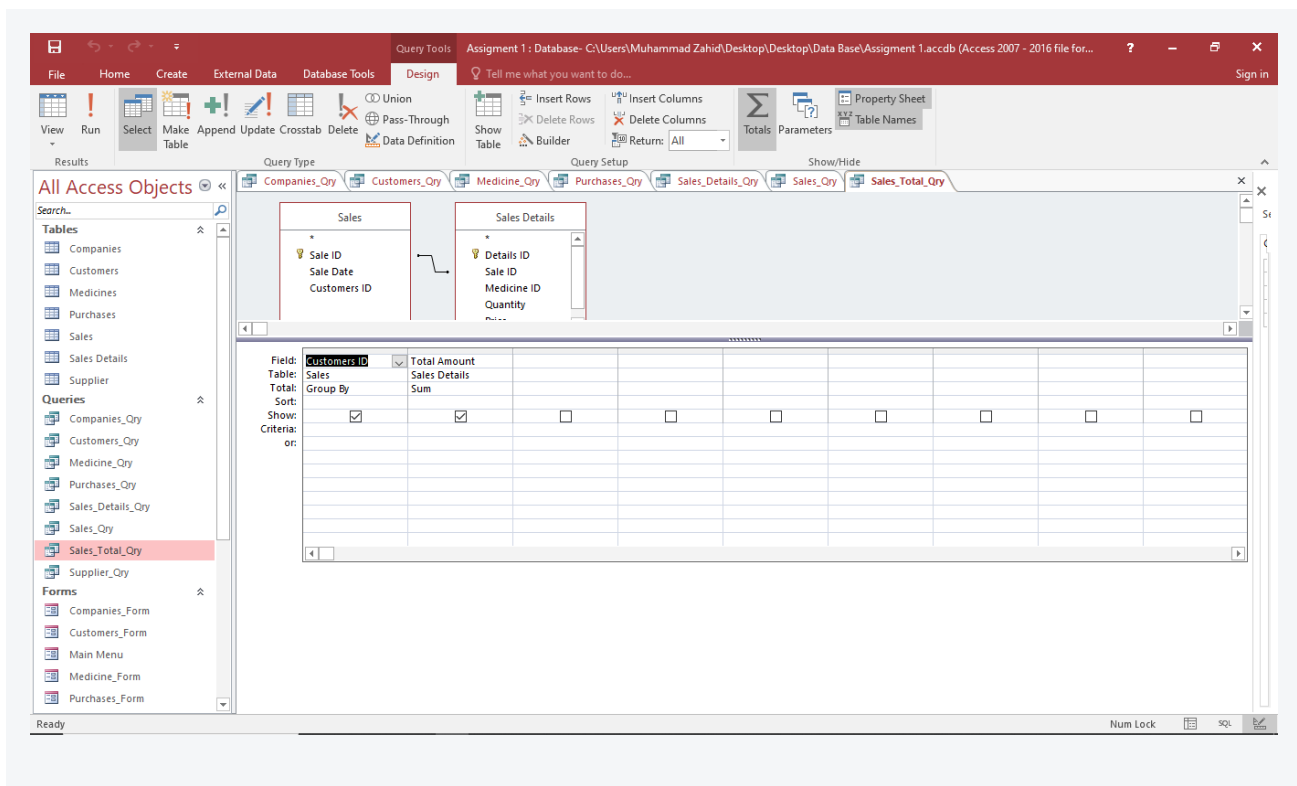


Figure 8: Sales Total Aggregation SQL Query View

5.2 Purchases Calculation Query (Purchases_Qry)

SQL Code:

```
SELECT Purchases.[Purchase Date], Supplier.[Supplier Name], Purchases.Quantity,
Purchases.[Cost Price], [Quantity]*[Cost Price] AS [Total Amount] FROM Supplier INNER
JOIN Purchases ON Supplier.[Supplier ID] = Purchases.[Supplier ID];
```

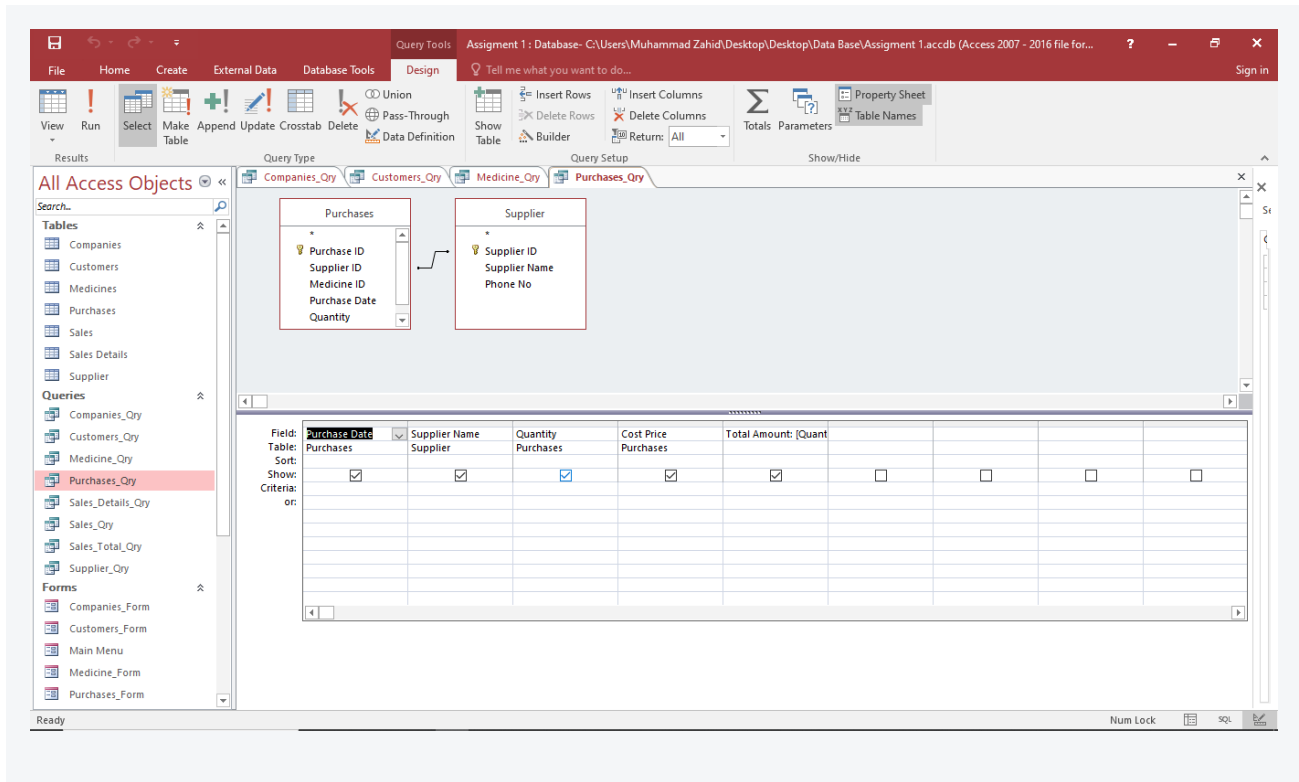


Figure 9: Purchases Calculation Query SQL Execution

CHAPTER 6: USER INTERFACE & FORMS DESIGN

The graphical interface consists of 8 user-friendly forms for daily operations:

6.1 Main Switchboard Form

The screenshot displays the 'Main Menu' window of a 'Pharmacy Management System'. The title bar shows 'Main Menu' and standard window controls. The header bar is dark green with a home icon, the system name, a subtitle 'Manages Medicines , Sales , Customers , Companies , Purchases and Suppliers', and the date/time '26-Aug-26 11:45:31 am'. Below the header is a navigation bar with six buttons: 'Medicine', 'Sales', 'Customers', 'Companies', 'Purchases', and 'Suppliers'. The main content area is light green and contains a form for 'Medicine ID' (1), 'Medicine Name' (Panadol 500mg), 'Company Name' (GSK Pharma), 'Batch No' (B123), 'Expiry Date' (01/08/2026), 'Purchase Price' (Rs5.00), 'Sale Price' (Rs8.00), and 'Stock Quantity' (150). To the right of the form is a 'Quick Actions' section with a 'Search Box' and two 'Open Report' buttons. Below the form are navigation buttons: 'First', 'Previous', 'Next', 'Last', 'Search', 'Save', and 'Delete'. At the bottom, a status bar shows 'Record: 1 of 26', 'Unfiltered', and a search input field.

Main Menu

Pharmacy Management System

Manages Medicines , Sales , Customers , Companies , Purchases and Suppliers

26-Aug-26
11:45:31 am

Medicine Sales Customers Companies Purchases Suppliers

Medicine ID: 1
Medicine Name: Panadol 500mg
Company Name: GSK Pharma
Batch No: B123
Expiry Date: 01/08/2026
Purchase Price: Rs5.00
Sale Price: Rs8.00
Stock Quantity: 150

Quick Actions

Search Box

Open Report

Open Report

First Previous Next Last Search Save Delete

Record: 1 of 26 Unfiltered Search

Figure 10: Main Menu / Navigation Switchboard Form

6.2 Sales Invoicing Master-Detail Form

The screenshot displays the Microsoft Access application window titled "Assignment 1: Database- C:\Users\Muhammad Zahid\Desktop\Desktop\Data Base\Assignment 1.accdb (Access 2007 - 2016 file format) - Access". The interface includes a ribbon with tabs: File, Home, Create, External Data, Database Tools, and Tell me what you want to do... The left-hand pane shows the "All Access Objects" list, categorized into Forms, Reports, and Queries. The "Sales_Form" is currently selected under the Forms category. The main window displays the "Sales_Form" in Form View, which is a master-detail form. The master form, "Sales_Form", contains fields for "Sale ID" (value: 1), "Sale Date" (value: 01/06/2026), and "Name" (value: Ali Khan). It also features navigation buttons: First, Previous, Next, Last, Search, New, and Delete. The detail form, "Subform_SalesDetails", is embedded within the master form and displays fields for "Sale ID" (value: 1), "Medicine Name" (value: Augmentin 6), "Quantity" (value: 7), "Price" (value: Rs5.00), and "Total Amount" (value: Rs35.00). It also has navigation buttons: Save, Search, New, and Delete. The status bar at the bottom indicates "Form View" and "Records: 1 of 25".

Figure 11: Master-Detail Sales Invoicing Form (Sales_Form & Subform_SalesDetails)

6.3 Medicine Stock Entry Form

The screenshot displays the Microsoft Access application window titled "Assignment 1 : Database- C:\Users\Muhammad Zahid\Desktop\Desktop\Data Base\Assignment 1.accdb (Access 2007 - 2016 file format) - Access". The "Medicine_Form" is open, showing a data entry form with the following fields and values:

Field	Value
Medicine ID	
Medicine Name	Panadol 500mg
Company Name	GSK Pharma
Batch No	B123
Expiry Date	01/08/2026
Purchase Price	Rs5.00
Sale Price	Rs8.00
Stock Quantity	150

Navigation buttons at the bottom include: First, Previous, Next, Last, Search, Save, and Delete. A "Quick Actions" box on the right contains a "Search Box" and two "Open Report" buttons. The status bar at the bottom indicates "Records: 1 of 26" and "Unfiltered".

Figure 12: Medicine Inventory Management Form

6.4 Customer Registration Form

The screenshot shows the Microsoft Access interface with the 'Customers_Form' open. The ribbon at the top includes 'File', 'Home', 'Create', 'External Data', 'Database Tools', and 'Tell me what you want to do...'. The 'Home' ribbon is active, showing options for 'View', 'Clipboard', 'Filter', 'Sort & Filter', 'Records', 'Find', and 'Text Formatting'. The 'All Access Objects' task pane on the left lists various objects, with 'Customers_Form' selected under the 'Forms' category. The form itself has a light blue header and contains the following fields and controls:

- Customers ID:** A text box containing the value '1'.
- Customers Name:** A text box containing the value 'Ali Khan'.
- Phone:** A text box containing the value '0312-4568781'.
- Address:** A text box containing the value 'Saddar Karachi'.
- Buttons:** Two 'Open Report' buttons are positioned to the right of the 'Customers Name' and 'Phone' fields. At the bottom of the form, there is a row of navigation buttons: 'First', 'Previous', 'Next', 'Last', 'Search', 'Save', and 'Delete'.

The status bar at the bottom indicates 'Form View' and 'Records: 1 of 1'.

Figure 13: Customer Directory & Registration Form

CHAPTER 7: DYNAMIC REPORTS & OUTPUTS

The database generates 7 automated printable reports:

7.1 Daily Sales Summary Report

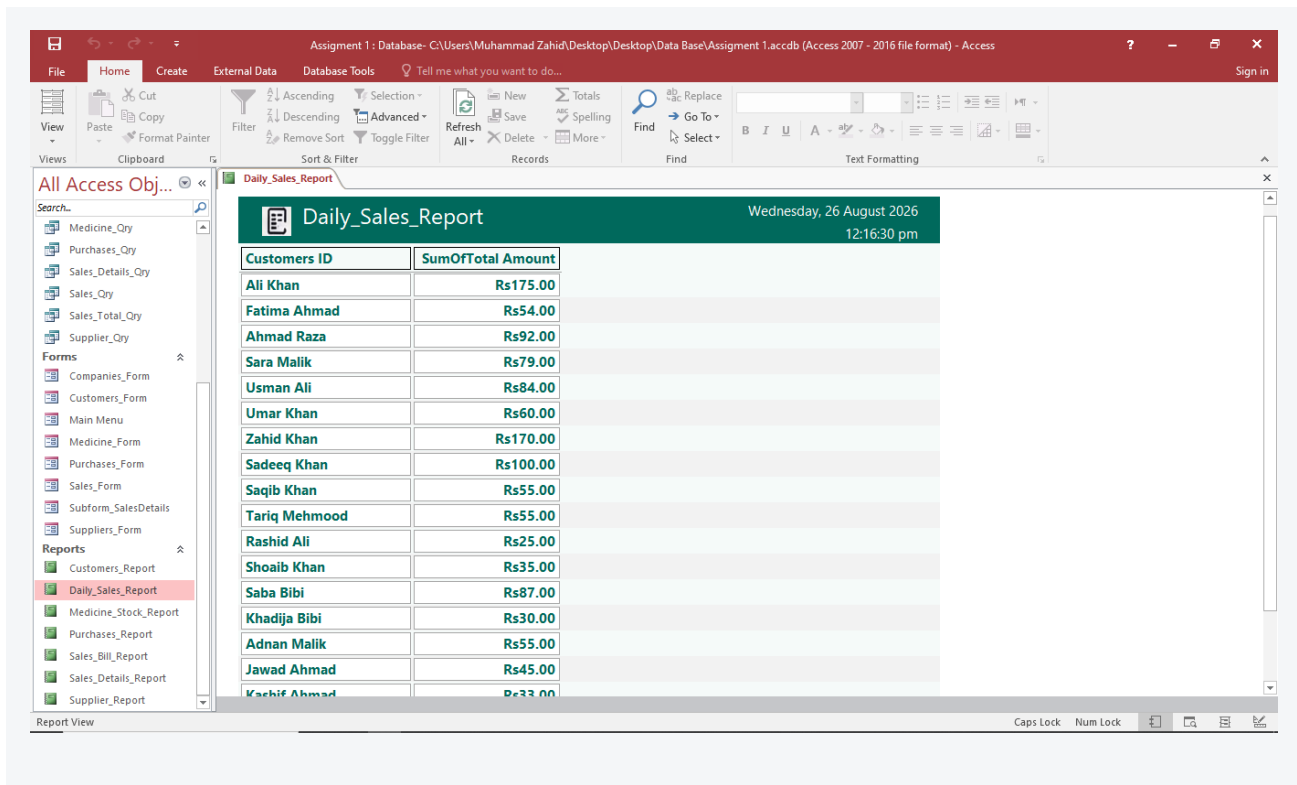


Figure 14: Daily Sales Summary Financial Report

7.2 Medicine Stock Report

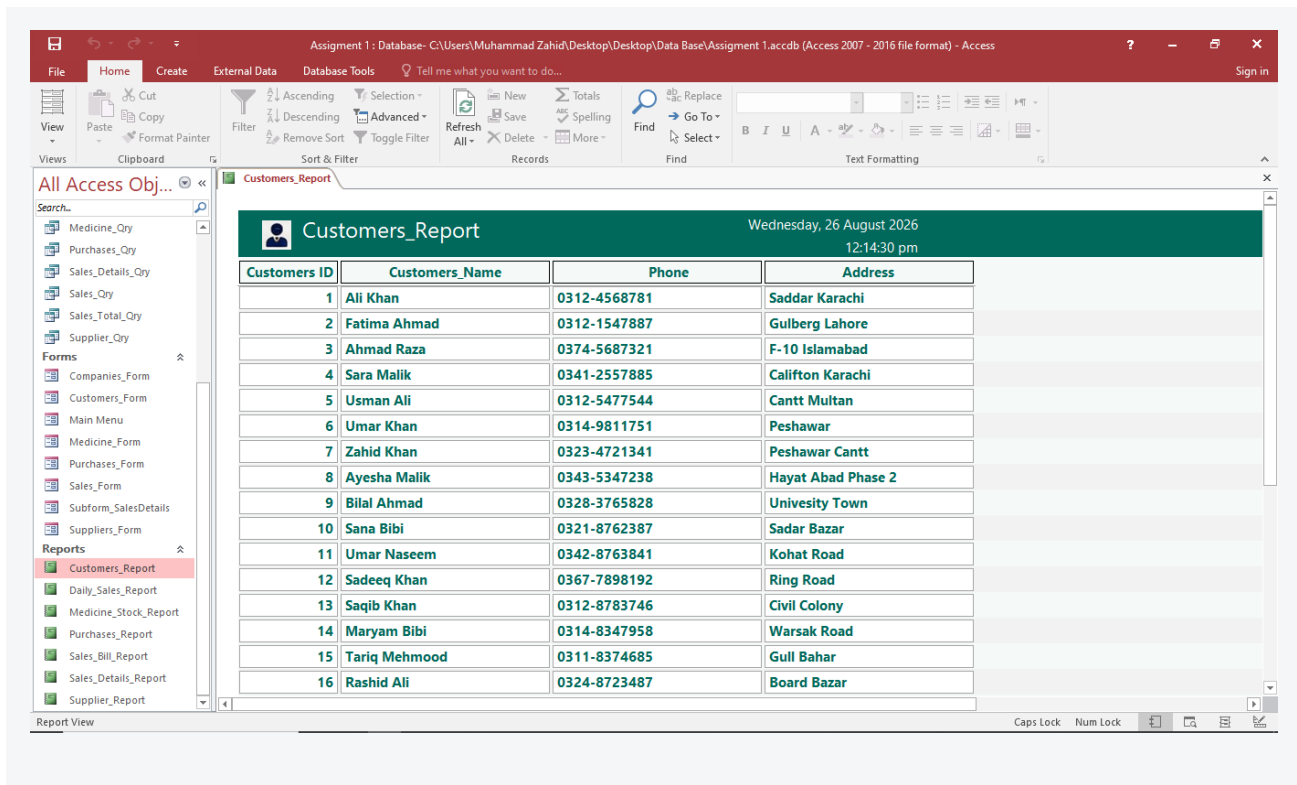
Assignment 1: Database- C:\Users\Muhammad Zahid\Desktop\Desktop\Data Base\Assignment 1.accdb (Access 2007 - 2016 file format) - Access

Wednesday, 26 August 2026
12:15:25 pm

Medicine ID	Medicine Name	Company ID	Batch No	Expiry Date	Purchase Price	Sale Price	Stock Quant
1	Panadol 500mg	GSK Pharma	B123	01/08/2026	Rs5.00	Rs8.00	1
2	Brufen 400mg	Getz Pharma	P456	09/07/2026	Rs12.00	Rs18.00	
3	Flagy 400mg	Pfizer Pakistan	F789	04/09/2026	Rs8.00	Rs12.00	
4	Augmentin 625mg	Martin Dow	A101	12/08/2026	Rs45.00	Rs65.00	
5	Panadol Extra	Getz Pharma	D206	23/10/2026	Rs4.00	Rs9.00	1
6	Omni dol	GSK Pharma	S201	27/06/2026	Rs10.00	Rs15.00	
7	Calpol Syrup	Martin Dow	A202	01/05/2026	Rs8.00	Rs14.00	
8	Zantac 150mg	Merck Pakistan	C404	02/05/2026	Rs9.00	Rs12.00	
9	Cifixime 400mg	Pfizer Pakistan	Z503	03/05/2026	Rs10.00	Rs16.00	
10	Losec 20mg	Barrett Hengston	C604	04/05/2026	Rs25.00	Rs38.00	
11	Ventolin Inhaler	Abbot Labs	I804	19/05/2026	Rs12.00	Rs18.00	
12	Betadine Solution	Dawn Pharma	L705	07/05/2026	Rs45.00	Rs56.00	
13	Buscopan 10mg	Highnoon	V806	18/06/2026	Rs30.00	Rs40.00	
14	Arinac Forte	Searle Pakistan	B908	20/06/2026	Rs10.00	Rs19.00	
15	Neurobion	Sanofi Aventis	B008	24/05/2026	Rs60.00	Rs75.00	
16	Flovit Tablets	Weyth Pakistan	A108	16/06/2026	Rs45.00	Rs56.00	
17	Calcium Sandoz	Roche Pakistan	N206	14/05/2026	Rs32.00	Rs45.00	

Figure 15: Medicine Stock Ledger & Expiry Report

7.3 Customer Sales Bill Receipt



The screenshot shows the Microsoft Access application window titled "Assignment 1: Database- C:\Users\Muhammad Zahid\Desktop\Desktop\Data Base\Assignment 1.accdb (Access 2007 - 2016 file format) - Access". The ribbon includes File, Home, Create, External Data, Database Tools, and Tell me what you want to do... The navigation pane on the left shows the "Customers_Report" report selected under the "Reports" section. The main area displays the "Customers_Report" report, which is a table with 4 columns: Customers ID, Customers_Name, Phone, and Address. The report is dated Wednesday, 26 August 2026, at 12:14:30 pm. The table contains 16 rows of customer data.

Customers ID	Customers_Name	Phone	Address
1	Ali Khan	0312-4568781	Saddar Karachi
2	Fatima Ahmad	0312-1547887	Gulberg Lahore
3	Ahmad Raza	0374-5687321	F-10 Islamabad
4	Sara Malik	0341-2557885	Califton Karachi
5	Usman Ali	0312-5477544	Cantt Multan
6	Umar Khan	0314-9811751	Peshawar
7	Zahid Khan	0323-4721341	Peshawar Cantt
8	Ayesha Malik	0343-5347238	Hayat Abad Phase 2
9	Bilal Ahmad	0328-3765828	Univesity Town
10	Sana Bibi	0321-8762387	Sadar Bazar
11	Umar Naseem	0342-8763841	Kohat Road
12	Sadeeq Khan	0367-7898192	Ring Road
13	Saqib Khan	0312-8783746	Civil Colony
14	Maryam Bibi	0314-8347958	Warsak Road
15	Tariq Mehmood	0311-8374685	Gull Bahar
16	Rashid Ali	0324-8723487	Board Bazar

Figure 16: Itemized Customer Billing Receipt / Invoice Report

CHAPTER 8: CONCLUSION & FUTURE SCOPE

The Pharmacy Management System successfully fulfills all requirements of the DIT final project curriculum. It eliminates manual errors, speeds up invoicing, and gives pharmacy owners real-time control over stock.